EXPERIMENT 6

INTER PROCESS COMMUNICATION

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/ipc.h>
#include <sys/shm.h>

#define SIZE 1024

int main() {
    int shmid;
    key_t key = 1234;
    char *shm;

    // Create or access shared memory
    shmid = shmget(key, SIZE, IPC_CREAT | 0666);
    if (shmid < 0) {
        perror("shmget failed");
        exit(1);
    }

    // Attach shared memory
    shm = (char*) shmat(shmid, NULL, 0);
    if (shm == (char *) -1) {
        perror("shmat failed");
```

```
        exit(1);
    }

    int choice;

    printf("\n1. Write to Shared Memory\n2. Read from Shared Memory\nEnter
choice: ");
    scanf("%d", &choice);
    getchar(); // clear buffer

    if (choice == 1) {
        printf("Enter data: ");
        fgets(shm, SIZE, stdin);
        printf("Data written into shared memory: %s\n", shm);
    }
    else if (choice == 2) {
        printf("Data read from shared memory: %s\n", shm);
    }
    else {
        printf("Invalid choice\n");
    }

    // Detach shared memory
    shmdt(shm);

    return 0;
}
```

OUTPUT

Run 1 (Writing Data):

1. Write to Shared Memory
2. Read from Shared Memory

Enter choice: 1

Enter data: Hello IPC

Data written into shared memory: Hello IPC

Run 2 (Reading Data):

1. Write to Shared Memory
2. Read from Shared Memory

Enter choice: 2

Data read from shared memory: Hello IPC